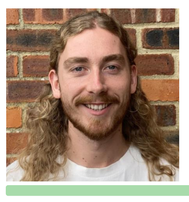
# Using interface observations in weakly-coupled data assimilation

**Tom Hill**[1, 2, 3]

Yumeng Chen[2, 3], Amos Lawless[2, 3], Nancy Nichols[2, 3]
Dan Lea[1], Tsz Yan Leung[4], Jo Waller[1, 2]


## Introduction

Initialising model predictions using **data assimilation requires observation operators** which map from model to observation space. **Incremental 4D-Var** data assimilation **requires linearising these operators**.

**Interface observations** are any observations **sensitive to more than one Earth system component**. For example, radiation from the ocean surface interacts with the atmosphere before hitting satellite sensors.

In many operational systems, the **atmosphere and ocean are weakly-coupled** (Figure 1). Linearised interface observation operators are not straightforward to use in weakly-coupled incremental 4D-Var.

We propose a way to use interface observations in these systems, which introduces an extra error source. To account for this error source, we derive a modification to the error covariance model which weights observation contributions during assimilation. We evaluate this approach in a simple coupled model.

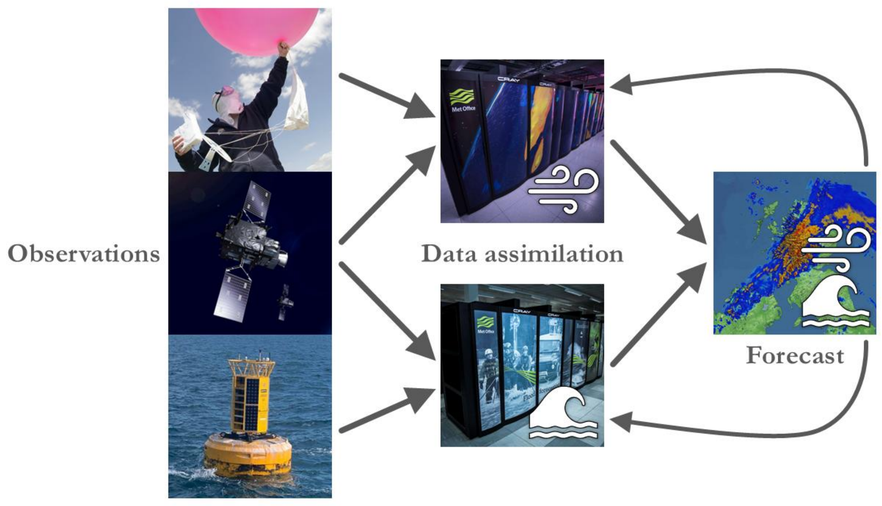



Figure 1: Schematic representation of weakly-coupled ocean-atmosphere data assimilation. The forecast model is coupled, but the assimilation optimisations are independent. Images from Met Office asset library.

## Theory

Observations, Observation operator, Innovations, Coupled background

$$\boldsymbol{d} = \boldsymbol{y} - H(\boldsymbol{x}_b)$$

Background, Analysis, Analysis increment

$$\boldsymbol{x}_a = \boldsymbol{x}_b + \delta\boldsymbol{x}_a$$

**Strongly-coupled**: Linearised operator $\mathbf{H}$ shows how adjustments $\delta\boldsymbol{x}$ affect fit to $\boldsymbol{y}$.
**Weakly-coupled**: Adjustments $\delta\boldsymbol{x}^{atmosphere}$ and $\delta\boldsymbol{x}^{ocean}$ are independent.

**Problem 1**: Independent adjustments mean coupled $\mathbf{H}$ cannot be used.
**Solution 1**: Use components from $\mathbf{H} = [\mathbf{H}^{at.} \quad \mathbf{H}^{oc.}]$ for respective adjustments.

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{at.}^{at.} & \mathbf{0} \\ \mathbf{H}_{in.}^{at.} & \mathbf{H}_{in.}^{oc.} \\ \mathbf{0} & \mathbf{H}_{oc.}^{oc.} \end{bmatrix}$$

Atmosphere-only observations; Interface observations; Ocean-only observations; Coupled state vector size

$$\mathbf{H}^{at.} = \begin{bmatrix} \mathbf{H}_{at.}^{at.} \\ \mathbf{H}_{in.}^{at.} \end{bmatrix}$$

Atmosphere state vector size

$$\mathbf{H}^{oc.} = \begin{bmatrix} \mathbf{H}_{in.}^{oc.} \\ \mathbf{H}_{oc.}^{oc.} \end{bmatrix}$$

Ocean state vector size

**Problem 2**: Each adjustment lacks part of full operator, which introduces error.
**Solution 2**: Modify observation error covariance model $\mathbf{R}$ to account for the error.

The derivation can be interpreted as matching some terms of the analysis equation (Kalman gain) for the weakly-coupled analyses to their equivalents in the strongly-coupled analysis. It uses components of the background error covariance model $\mathbf{B}$:

$$\mathbf{R}^{at.} = \mathbf{H}^{oc.}\mathbf{B}^{oc.}\mathbf{H}^{oc.T} \qquad \mathbf{R}^{oc.} = \mathbf{H}^{at.}\mathbf{B}^{at.}\mathbf{H}^{at.T}$$

The above is 3D-Var. A 4D-Var version requires coupled linear forecast and background error covariance models which, along with observation error correlations, are not fully implemented in operational systems. An alternative "basic" modification encodes these operational limitations by ignoring the related terms.

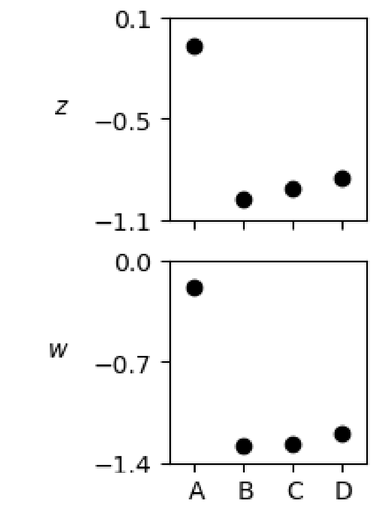



Figure 2: **Single observation experiment** analysis increments for an atmosphere (top) and ocean (bottom) state variable.

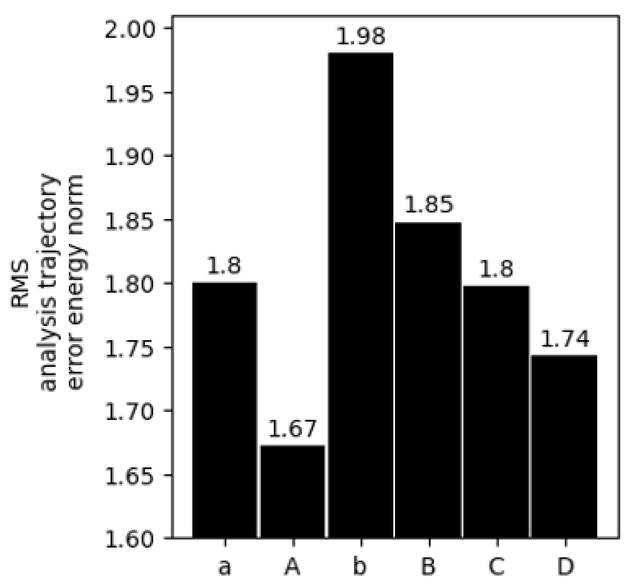



Figure 3: **Cycling experiment** summary errors.



Figure 4: ESA's Sentinel 3 carrying a radar altimeter and microwave calibration instrument (ESA–Pierre Carril, 2018). Ocean data assimilation systems assimilate sea surface height retrievals using this data, but an operator which treats the radar signal as an interface observation could provide an alternative, more direct mode of assimilation.

## Toy model experiments

We want to show that:

- interface observations can add value to a weakly-coupled system;
- modifying the observation error covariance model improves this further;
- an improvement is still possible even when the modification ignores components that are not available in current operational systems (coupled linear forecast and background error covariance models).

**Nonlinear forecast model**: The model of Molteni et al. (1993) is comprised of five coupled ODEs (three atmosphere, two ocean). Chosen parameter values instigate chaotic behaviour and produce an ocean which oscillates slower than the atmosphere.

**Single observation experiment** with a nonlinear coupled observation operator:
A. strongly-coupled incremental 4D-Var;
B. weakly-coupled incremental 4D-Var;
C. weakly-coupled with "basic" observation error (co)variance modification;
D. weakly-coupled with "complete" observation error (co)variance modification.
Figure 2 shows analysis increments for one atmosphere and one ocean state variable with each system. Both modifications move the analyses towards the strongly-coupled analysis, the complete modification more so. Results vary between state variables and with different randomly sampled observations and background states, but the case shown is typical.

**Cycling experiment** with multiple interface observations: In addition to systems A-D, three additional systems are run for sanity checks: no assimilation, and strongly- and weakly-coupled with interface observations excluded (a and b). Summary statistics are computed for each – a time average of an error norm of the trajectory with respect to truth. All systems are better than no assimilation, and analyses always improve upon backgrounds. Figure 3 shows the remaining results. Interface observations always add value, even in the unmodified weakly-coupled system. Strongly-coupled is always better than weakly-coupled, but the modifications provide an improvement, the basic modification less so.

## Conclusions and next steps

Through experiments with a nonlinear coupled toy model, we have shown that modifying the observation error covariance model according to linear theory to account for the splitting up of interface observation operator sensitivities can improve the performance of weakly-coupled incremental 4D-Var.

We have shown that this is effective when certain terms are ignored, but the approach is still infeasible in operational systems as it requires explicit matrix computations. We will investigate ways to approximate the modification further.

We will implement and assess the method with a different toy model in the Joint Effort for Data assimilation Integration (JEDI) system, both as a step towards implementing it in a research mode of the Met Office system and to allow investigation of real-world interface observations (an example is given in Figure 4).

## References


F. Molteni, L. Ferranti, T. N. Palmer, and P. Viterbo, A Dynamical Interpretation of the Global Response to Equatorial pacific SST Anomalies, Journal of Climate 6 (1993), no. 5, 777–795.



[1] Met Office [2] National Centre for Earth Observation, Natural Environment Research Council [3] University of Reading [4] ECMWF


AFESP Advancing the Frontiers of Earth System Prediction